

Exercise 1 (Negligible Functions). Prove or disprove that the following functions are negligible. You can use that $f(n) = \frac{1}{2^n}$ is negligible.

1. $f_1(n) = \frac{1}{n^{20}}$

Solution. This function is not negligible since the inverse of $g_1(n) = n^{21}$ is smaller than $f_1(n)$ for $n > 1$: $f_1(n) = \frac{1}{n^{20}} > \frac{1}{n^{21}} = \frac{1}{g_1(n)}$.

2. $f_2(n) = \frac{2^n}{(2^n)^2}$

Solution. This function is negligible since $f_2(n) = \frac{2^n}{(2^n)^2} = \frac{1}{2^n}$.

3. $f_3(n) = \left| \frac{(-1)^n}{2^n} \right|$

Solution. This function is negligible since $f_3(n) = \left| \frac{(-1)^n}{2^n} \right| = \frac{|(-1)^n|}{|2^n|} = \frac{1}{2^n}$.

4. $f_4(n) = \frac{(-1)^n}{2^n}$

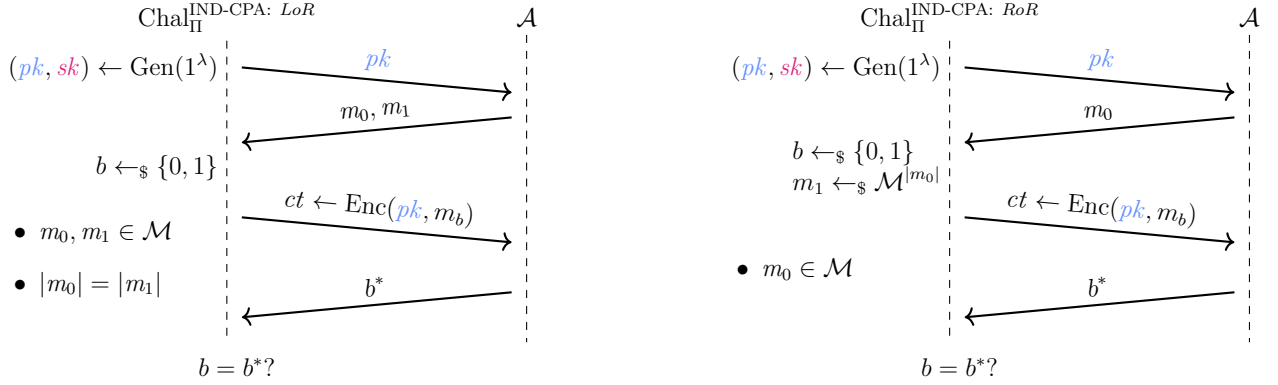
Solution. This function is negligible since $f_4(n) = \frac{(-1)^n}{2^n} = (-1)^n \cdot \frac{1}{2^n} \leq \frac{1}{2^n}$.

5. $f_5(n) = \frac{n^2 + n^7}{2^n}$

Solution. This function is negligible.

Formally, $f_5(n) = \frac{n^2 + n^7}{2^n} = (n^2 + n^7) \cdot \frac{1}{2^n} = q(n) \cdot h(n)$, where $q(n)$ is a polynomial and $h(n)$ negligible. Hence, $f_5(n)$ is negligible as well.

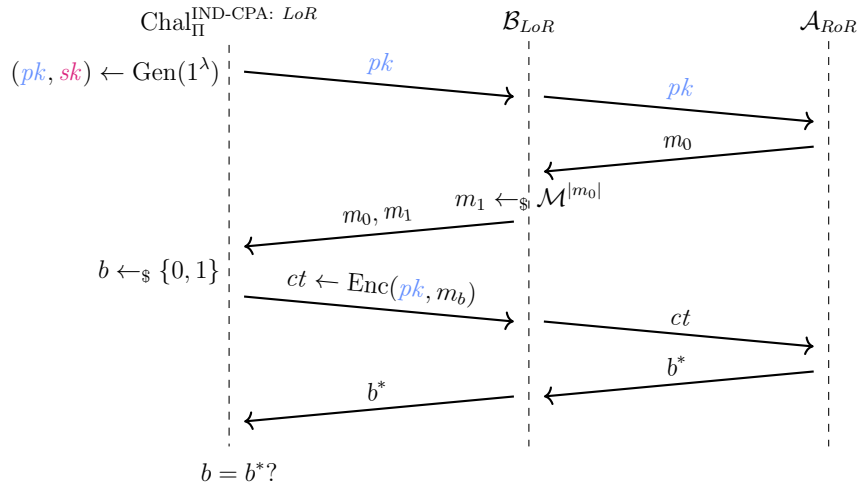
Exercise 2 (IND-CPA). Prove that Left-or-Right (LoR) IND-CPA (security game on the left) is equivalent to Real-or-Random (RoR) IND-CPA (security game on the right).



Solution. We show the statement in two steps.

$$(t, \epsilon)\text{-LoR} \implies (t', \epsilon')\text{-RoR}$$

Given an adversary $\mathcal{A}_{\text{RoR}}$ against RoR, a reduction $\mathcal{B}_{\text{LoR}}$ against LoR can win its game as follows:



The reduction is efficient, as all it does is forward queries and responses and sample one message per query.

The reduction simulates the RoR game for $\mathcal{A}_{\text{RoR}}$ as for every query m_0 the adversary $\mathcal{A}_{\text{RoR}}$ receives a ciphertext which is an encryption of m_0 or a randomly sampled message m_1 of the same length.

When $\mathcal{A}_{\text{RoR}}$ wins its game, it distinguishes between an encryption of m_0 and m_1 , which is exactly the condition for $\mathcal{B}_{\text{LoR}}$ to win its game as well.

Hence, we have $t' \approx t$ and $\epsilon' = \epsilon$.

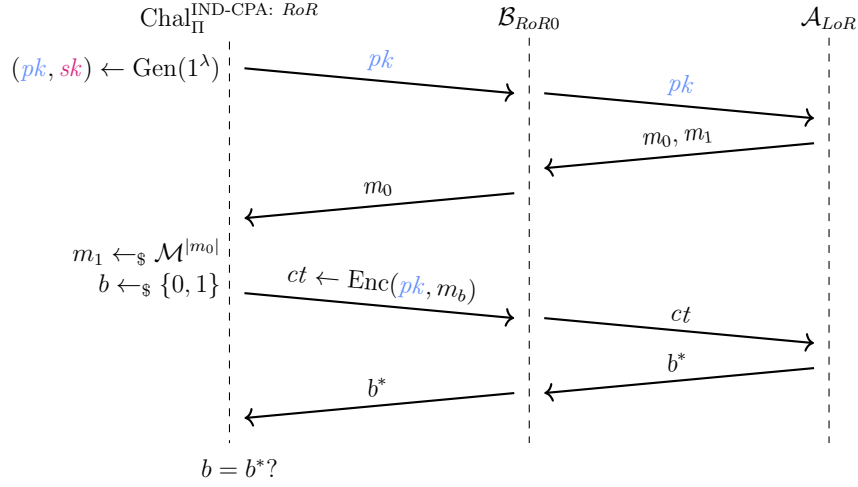
$$(t, \epsilon)\text{-RoR} \implies (t', \epsilon')\text{-LoR}$$

Here, we use a hybrid argument, similar to $\text{IND-CPA} \implies \text{Semantic Security}$ proof in the lecture. Consider these three games, which differ only in how the challenger creates the challenge ciphertext:

- Game 0: $\text{Enc}(pk, m_0)$
- Game 1: $\text{Enc}(pk, m_r)$ with $m_r \leftarrow_{\$} \mathcal{M}^{|m_0|}$

- Game 2: $\text{Enc}(pk, m_1)$.

We build a reduction $\mathcal{B}_{RoR0}$ that simulates Game 0 or 1 for an adversary $\mathcal{A}_{LoR}$ against LoR:



If $b = 0$, then $\mathcal{B}_{RoR0}$ simulates Game 0, and if $b = 1$, $\mathcal{B}_{RoR0}$ simulates Game 1. Similarly, we define a reduction $\mathcal{B}_{RoR1}$ that does the same as $\mathcal{B}_{RoR0}$ except that it forwards m_1 as challenge instead of m_0 . If $b = 0$, then $\mathcal{B}_{RoR1}$ simulates Game 1, and if $b = 1$, $\mathcal{B}_{RoR1}$ simulates Game 2.

We denote with ϵ_i the probability of the adversary $\mathcal{A}_{LoR}$ winning in Game i . By the triangle inequality we have

$$|\epsilon_0 - \epsilon_2| \leq |\epsilon_0 - \epsilon_1| + |\epsilon_1 - \epsilon_2|$$

By assumption the scheme is (t, ϵ) -RoR secure. We know that distinguishing between Games 0 and 2 is the same as LoR. Hence,

$$\epsilon = |\epsilon_0 - \epsilon_2| \leq |\epsilon_0 - \epsilon_1| + |\epsilon_1 - \epsilon_2| = 2\epsilon'.$$

Exercise 3 (Perfect Secrecy). Prove that perfect secrecy is impossible for PKE schemes.

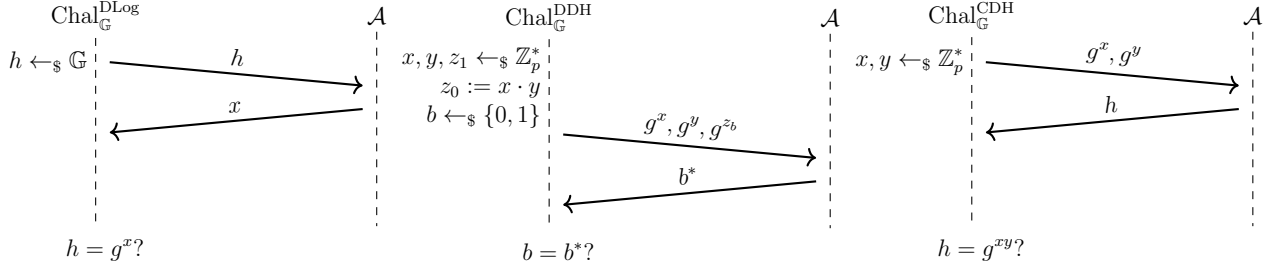
Solution. Recall that perfect secrecy requires that for every probability distribution for M , every message $m \in \mathcal{M}$, and every ciphertext $c \in \mathcal{C}$ for which $\Pr[C = c] > 0$:

$$\Pr[M = m | C = c] = \Pr[M = m].$$

An adversary receives a public key pk and a ciphertext c as input and brute-forces the secret key sk . In particular, for every sk , the adversary can encrypt every message with pk and check if the decryption with sk succeeds. Having found the matching sk , the adversary can decrypt c and learns the message.

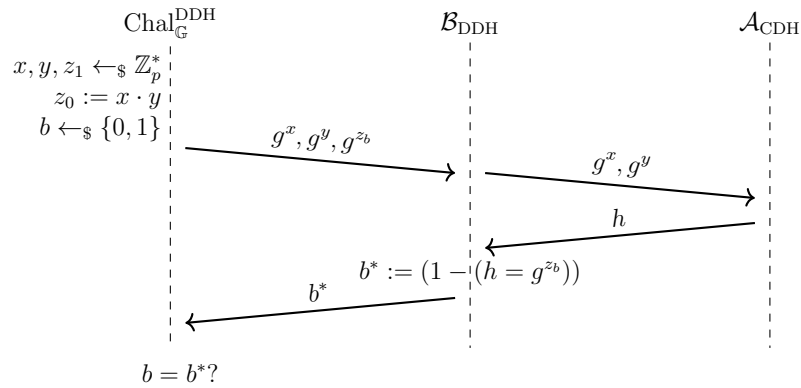
Furthermore, perfect secrecy requires that the key space is at least as big as the message space. Typically, PKE schemes have a message space $\mathcal{M} = \{0, 1\}^*$ and a limited key space.

Exercise 4 (DLog, DDH, and CDH). Let $\mathbb{G}$ be a group with generator g and modulus p . Recall the security games for DLog (left), DDH (middle), and CDH (right):



1. Show that if DDH is hard in $\mathbb{G}$, then so is CDH.

Solution. We show $\text{DDH} \implies \text{CDH}$ with a reduction:



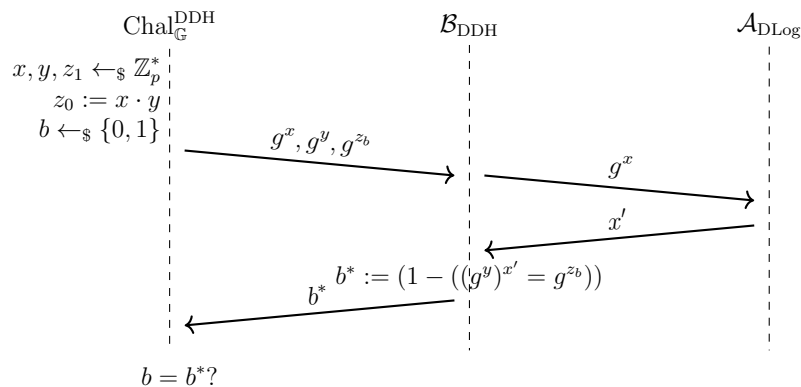
The reduction is efficient, as all it does is forward input and output and evaluate one boolean expression.

The reduction simulates the CDH game for $\mathcal{A}_{\text{CDH}}$ faithfully as it starts the adversary with two group elements.

When $\mathcal{A}_{\text{CDH}}$ wins its game, it has computed $h = g^{xy}$. The reduction compares this h to the g^{z_b} that it got as input. If both values are equivalent, the reduction knows that it has received a real DH triple and outputs $1 - (h = g^{z_b}) = 0$, and otherwise 1. Hence, the reduction $\mathcal{B}_{\text{DDH}}$ always wins its game when $\mathcal{A}_{\text{CDH}}$ wins its game.

2. Show that if DDH is hard in $\mathbb{G}$, then so is DLog.

Solution. We show $\text{DDH} \implies \text{DLog}$ with a reduction:



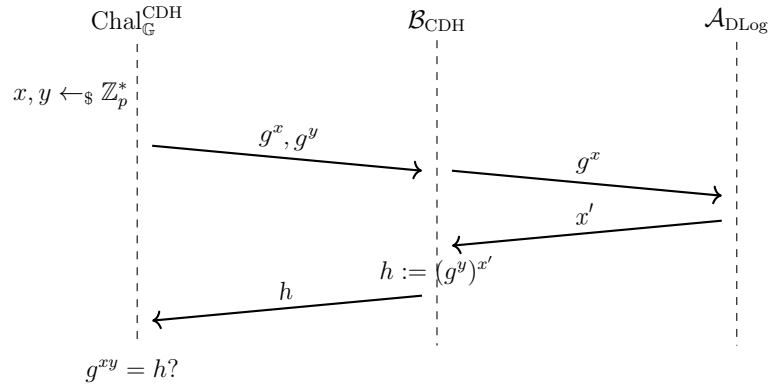
The reduction is efficient, as all it does is forward input and output and evaluate one boolean expression.

The reduction simulates the DLog game for $\mathcal{A}_{\text{DLog}}$ faithfully as it starts the adversary with one group element.

When $\mathcal{A}_{\text{DLog}}$ wins its game, it has computed x' with $g^x = g^{x'}$. The reduction compares $(g^y)^{x'}$ to the g^{z_b} that it got as input. If both values are equivalent, the reduction knows that it has received a real DH triple and outputs $1 - ((g^y)^{x'} = g^{z_b}) = 0$, and otherwise 1. Hence, the reduction $\mathcal{B}_{\text{DDH}}$ always wins its game when $\mathcal{A}_{\text{DLog}}$ wins its game.

3. Show that if CDH is hard in $\mathbb{G}$, then so is DLog.

Solution. We show $\text{CDH} \implies \text{DLog}$ with a reduction:



The reduction is efficient, as all it does is forward input and output and compute one exponentiation.

The reduction simulates the DLog game for $\mathcal{A}_{\text{DLog}}$ faithfully as it starts the adversary with one group element.

When $\mathcal{A}_{\text{DLog}}$ wins its game, it has computed x' with $g^x = g^{x'}$. The reduction computes $(g^y)^{x'}$ as its output, which is equivalent to g^{xy} exactly when x' is the discrete logarithm of g^x . Hence, the reduction $\mathcal{B}_{\text{CDH}}$ always wins its game when $\mathcal{A}_{\text{DLog}}$ wins its game.